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MACHINE LEARNING (25SC2107E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec no -5	**Multi-Class Match Result Prediction Using Multinomial Logistic Regression**

1.Abstract

Abstract

The project **"Multi-Class Match Result Prediction Using Multinomial Logistic Regression"** is a machine learning-based system developed to predict the outcome of **cricket matches**. The system collects historical and current cricket data such as **team performance, runs scored, wickets, venue, toss result, recent form, and player statistics** through a **Cricket Data API / Sports API**. The collected data is processed and prepared using **Python, Pandas, and NumPy**. **Google Colab** is used as the development environment for data preprocessing, model training, testing, and evaluation. Relevant features are selected and provided as input to the **Multinomial Logistic Regression** model using **Scikit-learn**, which predicts three possible outcomes: **Win, Loss, or No Result**. The model is evaluated using **accuracy, precision, recall, F1-score, and a confusion matrix** to measure its prediction performance.

The system is designed for **cricket fans, sports analysts, coaches, students, and sports organizations** who want to understand match patterns and obtain data-driven predictions. The trained machine learning model can be connected to a **Flask backend/API**, allowing users to provide match details and receive predictions from the trained model. **MySQL** can be used to store historical match data, user information, and prediction records, which can support future model improvement. Security features such as **JWT authentication, bcrypt password hashing, HTTPS/SSL, input validation, SQL injection prevention, Role-Based Access Control (RBAC), and environment variables** protect user and

application data. By combining **Cricket APIs, data preprocessing, Google Colab, Multinomial Logistic Regression, and a secure application**, the system provides users with an easy and data-driven way to analyze cricket performance and predict possible match outcomes.

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